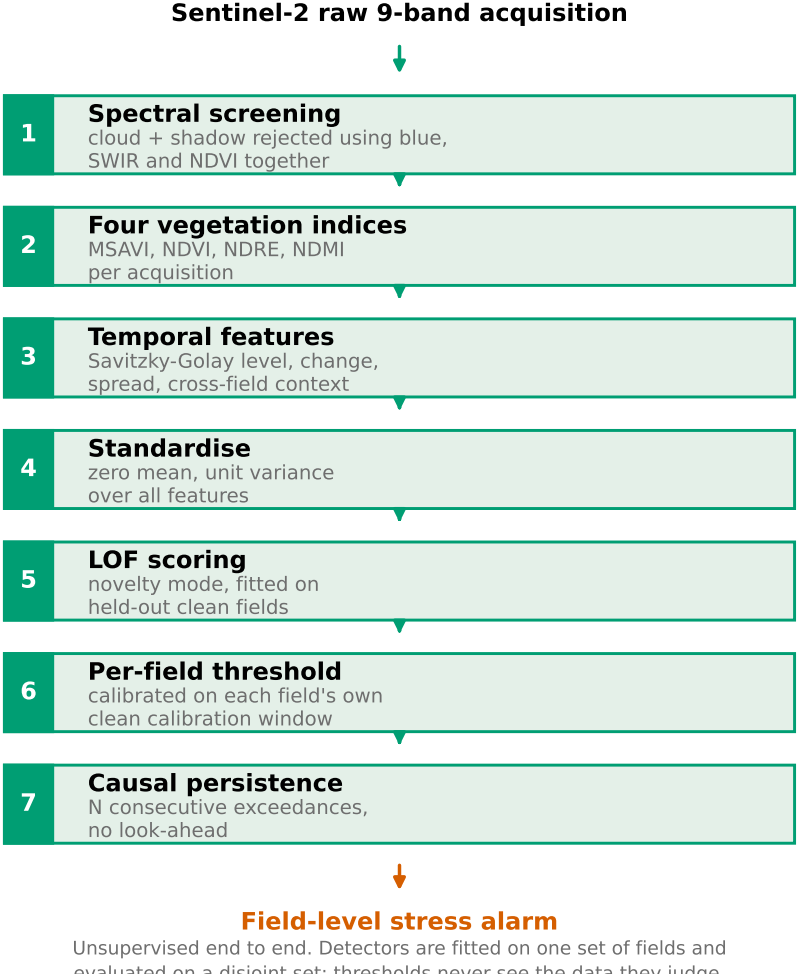
Sentinel-2 raw 9-band acquisition
1
Spectral screening
cloud + shadow rejected using blue, SWIR and NDVI together
2
Four vegetation indices
MSAVI, NDVI, NDRE, NDMI per acquisition
3
Temporal features
Savitzky-Golay level, change, spread, cross-field context
4
Standardise
zero mean, unit variance over all features
5
LOF scoring
novelty mode, fitted on held-out clean fields
6
Per-field threshold
calibrated on each field's own clean calibration window
7
Causal persistence
N consecutive exceedances, no look-ahead
Field-level stress alarm
Unsupervised end to end. Detectors are fitted on one set of fields and evaluated on a disjoint set; thresholds never see the data they judge.